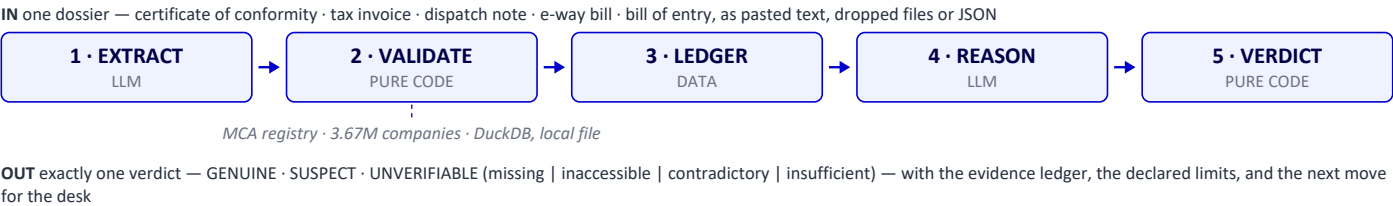


PROVENANCE — architecture note

Quessathon 2026 · Challenge 07 — Counterfeit Parts Verification · Team Mavericks

An agent that reads a part’s paper trail and decides, against records we do not control, whether the part is what its paperwork claims. Exactly one verdict — **GENUINE · SUSPECT · UNVERIFIABLE** — produced fully offline. We did not build a scenario catalogue; we built a **contradiction engine**: claims are routed to checks by their *type*, never by anticipated scenario, so an unseen case is a new combination of claims, not a missing branch.

1 System flow



STAGE	KIND	WHAT IT DOES
1 extract.py	LLM	Documents become typed assertions {entity, attribute, value, source_doc, date}. Document text is data , never instructions : instruction-like lines are stripped before the model sees them and logged as evidence of tampering.
2 validators.py	pure code	25 deterministic checks against the local MCA registry: GSTIN checksum / state / embedded PAN, CIN decode and registry cross-check, temporal sanity, e-way-bill validity vs distance, HSN vs part, port and mode of entry, BIS, TAC, lot-code grammar, cross-document drift, dossier reuse, and a successor lookup that goes hunting for a company no document names.
3 ledger.py	data	Every finding carries direction (genuine / suspect / neutral), strength (weak → dispositive), source tier (authoritative → derived → self-reported → heuristic) and dimension (identity, certification, provenance, custody).
4 reason.py	LLM	One benign and one malicious explanation per contradiction, adjudicated under fixed tier rules — then a second pass in which the agent attacks its own draft and may revise it. Switchable off live (rules-only mode).
5 verdict.py	pure code	One verdict and subtype, the work order, the action for each desk, and the case-aware limits.

Governance floors live in code, not in the prompt. Authoritative evidence beats any number of heuristics · tiers are never averaged · heuristics alone can never force SUSPECT · an absent registry record is a gap, not proof · identity evidence alone never certifies goods. No blended score is ever emitted. The self-critique may move a verdict only toward caution, and only when a non-heuristic contradiction exists — doubt manufactured from nothing is an abstention rate, not judgement. A malformed model answer falls back to the deterministic reasoner instead of reaching the verdict stage.

2 Dependencies

DEPENDENCY	WHAT IT IS	IF UNAVAILABLE
MCA registry	3.67M companies (data.gov.in Company Master Data, snapshot 2026-07-22) indexed in a local DuckDB file, mounted from a volume in deployment	The service still boots and reports the gap; dependent checks abstain and the verdict degrades to UNVERIFIABLE — never to a guess.
LLM (stages 1, 4)	Any Anthropic- or OpenAI-compatible endpoint behind one thin wrapper (demo: DeepSeek-chat, temperature 0)	Optional. With no key a deterministic parser and template reasoner take over; every result labels which engine ran.
Runtime	Python 3.12+, FastAPI, uvicorn, duckdb — the whole of requirements.txt	Required. No queue, no second database, no external service.
Frontend	React + Vite + TypeScript + Tailwind	Build-time only. The compiled bundle is committed, so a clone runs on Python alone.
BIS / GSTN portals	Deliberately <i>not</i> dependencies — both are captcha-gated	Modelled as a declared local stub and surfaced as the UNVERIFIABLE (inaccessible) subtype rather than silently skipped.
Seen-lots archive	An append-only local file of dossier fingerprints: the desk memory, and the only thing in the system that can see a <i>reused</i> dossier	Degrades to no finding at all on a read-only disk. Never consulted by a calibration run, so the offline eval stays deterministic.
Network	None in the core path — no CDN, no web fonts, no live lookups	Pipeline, UI, tests and calibration all run with the cable pulled.

3 Human hand-off points

- Briefing — the human decides what the agent sees.** Pasted paperwork, dropped files, a dossier JSON, or a bare CIN / GSTIN: nothing is auto-ingested. Before the run, *preflight* reports what the agent can see, the registry row it found, the checks it will run, and which document would unlock more.
- Ablation — the human can amputate the model.** The reviewer switches stage 4 off and re-runs the same dossier; the difference between the two answers is the model’s contribution, shown rather than claimed.
- The verdict is a recommendation, never an action.** The system never releases, quarantines, pays or emails. It issues the move for each desk — OEM brand protection, distributor incoming-goods QA (the primary user), showroom and service — and a person executes it.
- The abstention work order — the human sends it.** UNVERIFIABLE names its subtype, the single decisive missing artefact, an interim action and a low-confidence lean, then **drafts** the supplier request for a person to send. The lot waits on screen until it arrives. Nothing leaves the machine.
- Declared limits — the human is told what was not checked.** Every verdict ships a case-aware “what this system cannot determine” block: the physical part, image forensics, a byte-perfect cloned dossier, events after the registry snapshot, and intent.

Calibration. 15/15 exact (verdict + subtype) on our 15-case adversarial suite, identical on the mock and the live-LLM path at temperature 0 · 27% abstention · 100% accuracy when committed · rules-only scores 14/15, and that single divergence is the designed demonstration of what stage 4 adds · 306 seeded mutations of our own test data: 0 false GENUINES, 0 crashes. A design-verification claim, not a field-accuracy claim — no labelled corpus of counterfeit dossiers exists, and that absence is part of the problem. **Repository:** github.com/AaronHiresense/Provenance